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III/II CSE - R23 Regulations

BIG DATA ANALYTICS

UNIT I - UNDERSTANDING BIG DATA

UNIT I: big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.

INTRODUCTION TO BIG DATA

What is Big Data?

Big data refers to extremely large and diverse collections of structured, unstructured, and semi-structured data that continues to grow exponentially over time. These datasets are so huge and complex in volume, velocity, and variety, that traditional data management systems cannot store, process, and analyze them.

The amount and availability of data is growing rapidly, spurred on by digital technology advancements, such as connectivity, mobility, the Internet of Things (IoT), and artificial intelligence (AI). As data continues to expand and proliferate, new big data tools are emerging to help companies collect, process, and analyze data at the speed needed to gain the most value from it.

Big data describes large and diverse datasets that are huge in volume and also rapidly grow in size over time. Big data is used in machine learning, predictive modeling, and other advanced analytics to solve business problems and make informed decisions

The Vs of big data

Big data definitions may vary slightly, but it will always be described in terms of volume, velocity, and variety. These big data characteristics are often referred to as the "3 Vs of

- **Volume**

As its name suggests, the most common characteristic associated with big data is its high volume. This describes the enormous amount of data that is available for collection and produced from a variety of sources and devices on a continuous basis.

- **Velocity**

Big data velocity refers to the speed at which data is generated. Today, data is often produced in real time or near real time, and therefore, it must also be processed, accessed, and analyzed at the same rate to have any meaningful impact.

- **Variety**

Data is heterogeneous, meaning it can come from many different sources and can be structured, unstructured, or semi-structured. More traditional structured data (such as data in spreadsheets or relational databases) is now supplemented by unstructured text, images, audio, video files, or semi-structured formats like sensor data that can't be organized in a fixed data schema. big data" and were first defined by Gartner in 2001.

In addition to these three original Vs, three others that are often mentioned in relation to harnessing the power of big data: **veracity**, **variability**, and **value**.

- **Veracity:**

Big data can be messy, noisy, and error-prone, which makes it difficult to control the quality and accuracy of the data. Large datasets can be unwieldy and confusing, while smaller datasets could present an incomplete picture. The higher the veracity of the data, the more trustworthy it is.

- **Variability:**

The meaning of collected data is constantly changing, which can lead to inconsistency over time. These shifts include not only changes in context and interpretation but also data collection methods based on the information that companies want to capture and analyze.

- **Value:**

It's essential to determine the business value of the data you collect. Big data must contain the right data and then be effectively analyzed in order to yield insights that can help drive decision-making.

Sources of Big Data

These data come from many sources like

- **Social networking sites:** Facebook, Google, LinkedIn all these sites generate huge amount of data on a day to day basis as they have billions of users worldwide.
- **E-commerce site:** Sites like Amazon, Flipkart, Alibaba generates huge number of logs from which users buying trends can be traced.
- **Weather Station:** All the weather station and satellite gives very huge data which are stored and manipulated to forecast weather.
- **Telecom company:** Telecom giants like Airtel, Vodafone study the user trends and accordingly publish their plans and for this they store the data of its million users.
- **Share Market:** Stock exchange across the world generates huge amount of data through its daily transaction.

How does big data work?

The central concept of big data is that the more visibility you have into anything, the more effectively you can gain insights to make better decisions, uncover growth opportunities, and improve your business model.

Making big data work requires three main actions:

1. **Integration:**

Big data collects terabytes, and sometimes even petabytes, of raw data from many sources that must be received, processed, and transformed into the format that business users and analysts need to start analyzing it.

2. **Management:**

Big data needs big storage, whether in the cloud, on-premises, or both. Data must also be stored in whatever form required. It also needs to be processed and made available in real time. Increasingly, companies are turning to cloud solutions to take advantage of the unlimited compute and scalability.

3. **Analysis:**

The final step is analyzing and acting on big data—otherwise, the investment won't be worth it. Beyond exploring the data itself, it's also critical to communicate and share insights across the business in a way that everyone can understand. This includes using tools to create data visualizations like charts, graphs, and dashboards.

What is big data analytics?
Big data analytics is the process of collecting, examining, and analysing large amounts of data to discover market trends, insights, and patterns that can help companies make better business decisions. This information is available quickly and efficiently so that companies can be agile in crafting plans to maintain their competitive advantage.
Big data analytics is important because it helps companies leverage their data to identify opportunities for improvement and optimisation. Across different business segments, increasing efficiency leads to overall more intelligent operations, higher profits, and satisfied customers. Big data analytics helps companies reduce costs and develop better, customer-centric products and services.

Technologies such as business intelligence (BI) tools and systems help organisations take unstructured and structured data from multiple sources. Users (typically employees) input queries into these tools to understand business operations and performance. Big data analytics uses the four data analysis methods to uncover meaningful insights and derive solutions.

Types of big data analytics

Four main types of big data analytics support and inform different business decisions.

1. **Descriptive analytics**

Descriptive analytics refers to data that can be easily read and interpreted. This data helps create reports and visualise information that can detail company profits and sales.
Example: During the pandemic, a leading pharmaceutical company conducted data analysis on its offices and research labs. Descriptive analytics helped them identify consolidated unutilised spaces and departments, saving the company millions of pounds.

2. **Diagnostics analytics**

Diagnostics analytics helps companies understand why a problem occurred. Big data technologies and tools allow users to mine and recover data that helps dissect an issue and prevent it from happening in the future.
Example: An online retailer's sales have decreased even though customers continue to add items to their shopping carts. Diagnostics analytics helped to understand that the payment page was not working correctly for a few weeks.

3. **Predictive analytics**

Predictive analytics looks at past and present data to make predictions. With artificial intelligence (AI), machine learning, and data mining, users can analyse the data to predict market trends.
Example: In the manufacturing sector, companies can use algorithms based on historical data to predict if or when a piece of equipment will malfunction or break down.

4. **Prescriptive analytics**

Prescriptive analytics solves a problem, relying on AI and machine learning to gather and use data for risk management.
Example: Within the energy sector, utility companies, gas producers, and pipeline owners identify factors that affect the price of oil and gas to hedge risks.

Benefits of big data analytics
Incorporating big data analytics into a business or organisation has several advantages. These include:

Cost reduction: Big data can reduce costs in storing all business data in one place. Tracking analytics also helps companies find ways to work more efficiently to cut costs wherever possible.

Product development: Developing and marketing new products, services, or brands is much easier when based on data collected from customers' needs and wants. Big data analytics also helps businesses understand product viability and to keep up with trends.

Strategic business decisions: The ability to constantly analyse data helps businesses make better and faster decisions, such as cost and supply chain optimisation.

Customer experience: Data-driven algorithms help marketing efforts (targeted ads, for example) and increase customer satisfaction by delivering an enhanced customer experience.

Risk management: Businesses can identify risks by analysing data patterns and developing solutions for managing those risks.

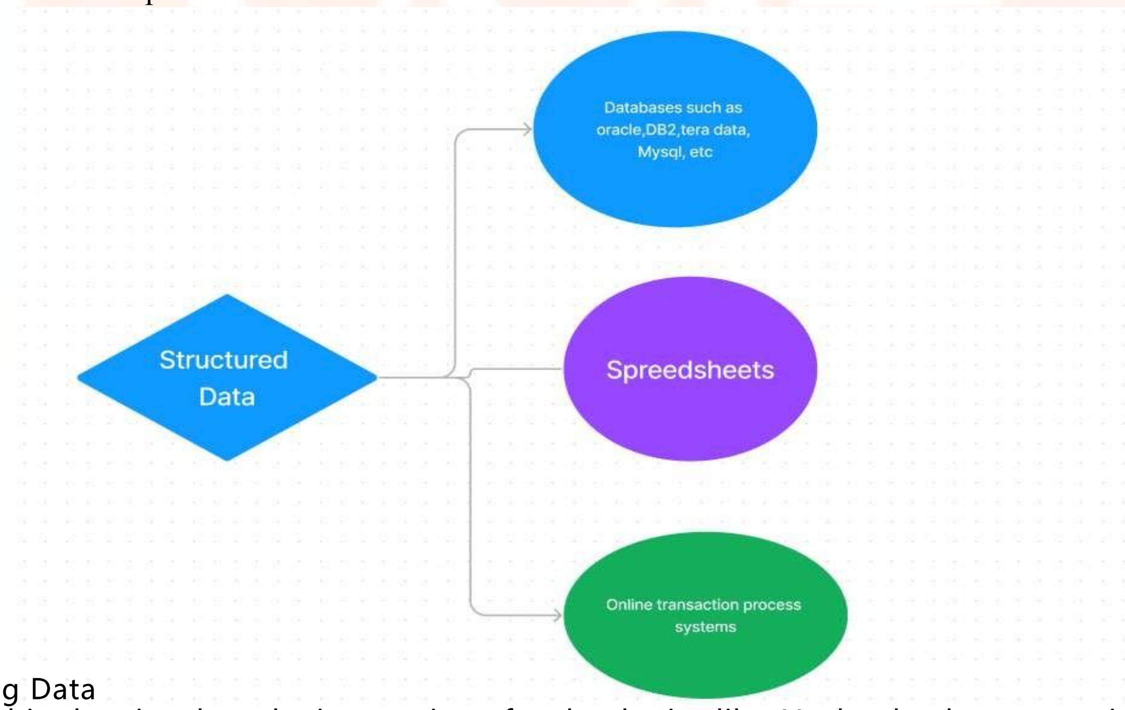
UNSTRUCTURED DATA

Types of Big Data

All data cannot be stored in the same way. The methods for data storage can be accurately evaluated after the type of data has been identified

1. Structured data

Structured data is data whose elements are addressable for effective analysis. It has been organized into a formatted repository that is typically a database. It concerns all data which can be stored in database in a table with rows and columns. They have relational keys and can easily be mapped into pre-designed fields. Today, those data are most processed in the development and simplest way to manage information. Example: Relational data.



Key Trends in Big Data

Convergence in big data involves the integration of technologies like AI, cloud, edge computing, and real-time processing to handle massive datasets more effectively. As of 2026, these trends emphasize speed, automation, and accessibility, driven by growing data volumes from IoT and AI applications.

1)AI and Machine Learning Integration: AI-native pipelines automate data processing, uncovering patterns in complex datasets for predictive analytics and decision-making. This symbiosis allows AI to train on big data while enhancing analysis of unstructured content via NLP and computer vision.

2)Real-Time and Streaming Analytics: Businesses now prioritize processing data as it arrives, using streaming tools like Kafka for instant insights in fraud detection and IoT maintenance. By 2025-2026, 75% of enterprise data is expected at the edge, reducing latency for time-sensitive operations.

3)Edge Computing and Cloud Synergy: Edge analytics processes data near sources like sensors, converging with cloud for scalability and low-latency AI deployment. This hybrid model supports real-time applications in manufacturing and smart cities while maintaining centralized governance.

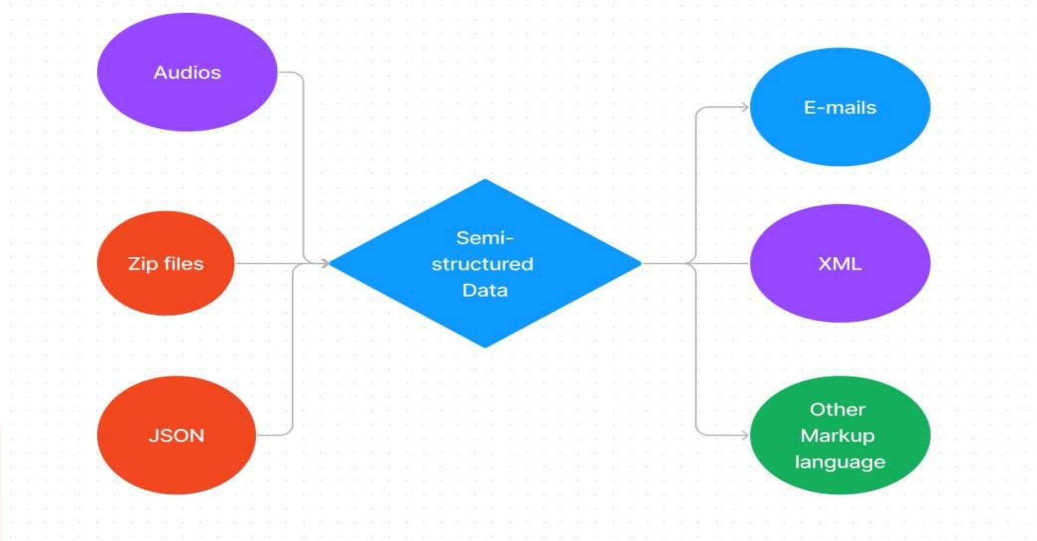
4)Data Governance and Privacy Regulations: As data grows, so do concerns about security and compliance. The convergence of big data with stricter governance frameworks ensures responsible and ethical use.

5}Volume, Velocity, Variety, and Veracity (the 4Vs): These foundational characteristics of big data are no longer treated in isolation. Advances in storage, processing, and analytics allow organizations to handle all four dimensions simultaneously.

6)Visualization and Business Intelligence Tools: Advanced dashboards and visualization platforms converge with big data to make insights accessible to decision-makers in real time.

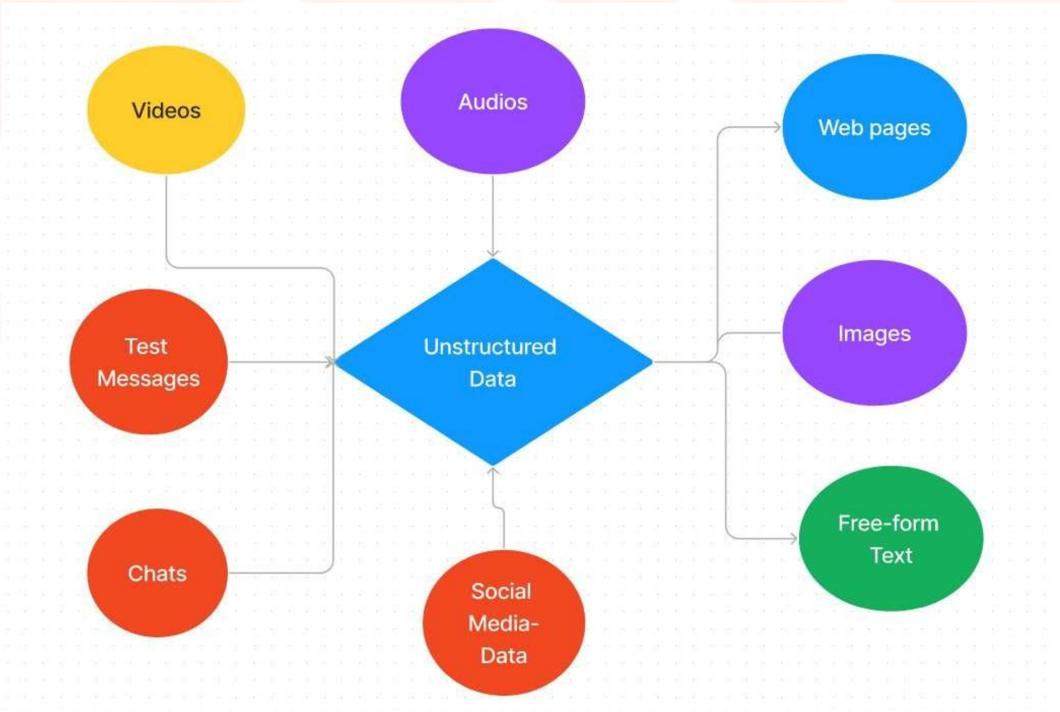
2. Semi-Structured data

Semi-structured data is information that does not reside in a relational database but that has some organizational properties that make it easier to analyze. With some processes, you can store them in the relation database (it could be very hard for some kind of semi-structured data), but Semi-structured exist to ease space. Example: XML data.



3. Unstructured data

Unstructured data is a data which is not organized in a predefined manner or does not have a predefined data model, thus it is not a good fit for a mainstream relational database. So for Unstructured data, there are alternative platforms for storing and managing, it is increasingly prevalent in IT systems and is used by organizations in a variety of business intelligence and analytics applications. Example: Word, PDF, Text, Media logs.



Unstructured data is the data which does not conforms to a data model and has no easily identifiable structure such that it can not be used by a computer program easily. Unstructured data is not organised in a pre-defined manner or does not have a pre-defined data model, thus it is not a good fit for a mainstream relational database.
From 80% to 90% of data generated and collected by organizations is unstructured, and its volumes are growing rapidly — many times faster than the rate of growth for structured databases.
Unstructured data stores contain a wealth of information that can be used to guide business decisions. However, unstructured data has historically been very difficult to analyze. With the help of AI and machine learning, new software tools are emerging that can search through vast quantities of it to uncover beneficial and actionable business intelligence.

Unstructured data vs. structured data
Let's take structured data first: it's usually stored in a relational database or RDBMS, and is sometimes referred to as relational data. It can be easily mapped into designated fields — for example, fields for zip codes, phone numbers, and credit cards. Data that conforms to RDBMS structure is easy to search, both with human-defined queries and with software.
Unstructured data, in contrast, doesn't fit into these sorts of pre-defined data models. It can't be stored in an RDBMS. And because it comes in so many formats, it's a real challenge for conventional software to ingest, process, and analyze. Simple content searches can be undertaken across textual unstructured data with the right tools.
Beyond that, the lack of consistent internal structure doesn't conform to what typical data mining systems can work with. As a result, companies have largely been unable to tap into value-laden data like customer interactions, rich media, and social network conversations. Robust tools for doing so are only now being developed and commercialized.

What are some examples of unstructured data?
Unstructured data can be created by people or generated by machines. Here are some examples of the *human-generated variety*:
- Email: Email message fields are unstructured and cannot be parsed by traditional analytics tools. That said, email metadata affords it some structure, and explains why email is sometimes considered semi-structured data.
- Text files: This category includes word processing documents, spreadsheets, presentations, email, and log files.
- Social media and websites: data from social networks like Twitter, LinkedIn, and Facebook, and websites such as Instagram, photo-sharing sites, and YouTube.
- Mobile and communications data: For this category, look no further than text messages, phone recordings, collaboration software, chat, and instant messaging.
- Media: This data includes digital photos, audio, and video files. Here are some examples of *unstructured data generated by machines:*
- Scientific data: This includes oil and gas surveys, space exploration, seismic imagery, and atmospheric data.
- Digital surveillance: This category features data like reconnaissance photos and videos.
- Satellite imagery: This data includes weather data, land forms, and military movements.

Characteristics of Unstructured Data:
- Data neither conforms to a data model nor has any structure.
- Data cannot be stored in the form of rows and columns as in Databases
- Data does not follow any semantic or rules
- Data lacks any particular format or sequence
- Data has no easily identifiable structure
- Due to lack of identifiable structure, it cannot used by computer programs easily

Sources of Unstructured Data:
- Web pages
- Images (JPEG, GIF, PNG, etc.)

- Videos
- Memos
- Reports
- Word documents and PowerPoint presentations
- Surveys

Advantages of Unstructured Data:

- Its supports the data which lacks a proper format or sequence
- The data is not constrained by a fixed schema
- Very Flexible due to absence of schema.
- Data is portable
- It is very scalable
- It can deal easily with the heterogeneity of sources.
- These types of data have a variety of business intelligence and analytics applications.

Disadvantages of Unstructured data:

- It is difficult to store and manage unstructured data due to lack of schema and structure
- Indexing the data is difficult and error prone due to unclear structure and not having pre-defined attributes. Due to which search results are not very accurate.
- Ensuring security to data is difficult task.

Problems faced in storing unstructured data:

- It requires a lot of storage space to store unstructured data.
- It is difficult to store videos, images, audios, etc.
- Due to unclear structure, operations like update, delete and search is very difficult.
- Storage cost is high as compared to structured data
- Indexing the unstructured data is difficult

Possible solution for storing Unstructured data:

- Unstructured data can be converted to easily manageable formats
- using Content addressable storage system (CAS) to store unstructured data. It stores data based on their metadata and a unique name is assigned to every object stored in it. The object is retrieved based on content not its location.
- Unstructured data can be stored in XML format.
- Unstructured data can be stored in RDBMS which supports BLOBs

Extracting information from unstructured Data:

- unstructured data do not have any structure. So it cannot easily interpreted by conventional algorithms. It is also difficult to tag and index unstructured data. So extracting information from them is tough job. Here are possible solutions:
- Taxonomies or classification of data helps in organising data in hierarchical structure. Which will make search process easy.
- Data can be stored in virtual repository and be automatically tagged. For example Documentum.
- Use of application platforms like XOLAP. XOLAP helps in extracting information from e-mails and XML based documents.
- Use of various data mining tools.

<u>BIG DATA INDUSTRY APPLICATIONS</u>

Here are some of the sectors where Big Data is actively used:

Ecommerce - Predicting customer trends and optimizing prices are a few of the ways e-commerce uses Big Data analytics
Marketing - Big Data analytics helps to drive high ROI marketing operations, which result in improved sales
Education - Used to develop new and improve existing courses based on market requirements
Healthcare - With the help of a patient's medical history, Big Data analytics is used to predict how likely they are to have health issues
Media and entertainment - Used to understand the demand of shows, movies, songs, and more to

deliver a personalized recommendation list to its users
Banking - Customer income and spending patterns help to predict the likelihood of choosing various banking offers, like loans and credit cards
Telecommunications - Used to forecast network capacity and improve customer experience
Government - Big Data analytics helps governments in law enforcement, among other things

WEB ANALYTICS

Web Analytics or Online Analytics refers to the analysis of quantifiable and measurable data of your website with the aim of understanding and optimizing the web usage.
Web Analytics is the methodological study of **online/offline** patterns and trends. It is a technique that you can employ to collect, measure, report, and analyze your website data. It is normally carried out to analyze the performance of a website and optimize its web usage.
web analytics used to track key metrics and analyze visitors' activity and traffic flow. It is a tactical approach to collect data and generate reports. It is an ongoing process that helps in attracting more traffic to a site and thereby, increasing the Return on Investment.

Web analytics focuses on various issues. For example,

- Detailed comparison of visitor data, and Affiliate or referral data.
- Website navigation patterns.
- The amount of traffic your website received over a specified period of time.
- Search engine data.

Web analytics improves online experience for your customers and elevates your business prospects. There are various Web Analytics tools available in the market. For example, Google Analytics, Kissmetrics, Optimizely, etc.
Importance of Web Analytics
Web Analytics needed to assess the success rate of a website and its associated business. Using Web Analytics, we can −

- Assess web content problems so that they can be rectified
- Have a clear perspective of website trends
- Monitor web traffic and user flow
- Demonstrate goals acquisition
- Figure out potential keywords
- Identify segments for improvement
- Find out referring sources

Web Analytics Process
The primary objective of carrying out Web Analytics is to optimize the website in order to provide better user experience. It provides a data-driven report to measure visitors' flow throughout the website.
Take a look at the following illustration. It depicts the process of web analytics.

- Set the business **goals**.
- To track the goal achievement, set the **Key Performance Indicators** (KPI).
- **Collect** correct and suitable data.
- To extract insights, **Analyze** data.
- Based on assumptions learned from the data analysis, **Test alternatives**.
- Based on either data analysis or website testing, **Implement insights**. **Types of Web Analytics**

There are two types of web analytics −

- **On-site** − It measures the users' behaviour once it is on the website. For example, measurement of your website performance.
- **Off-site** − It is the measurement and analysis irrespective of whether you own or maintain a website. For example, measurement of visibility, comments, potential audience, etc.

Metrics of Web Analytics
There are three basic metrics of web analytics −
Count
It is most basic metric of measurement. It is represented as a whole number or a fraction. For example,

- Number of visitors = 12999, Number of likes = 3060, etc.
- Total sales of merchandise = $54,396.18.

Ratio
It is typically a count divided by some other count. For example, Page views per visit.
Key Performance Indicator (KPI)
It depends upon the business type and strategy. KPI varies from one business to another.
Micro and macro Level Data Insights
Google Analytics gives you more insight data accurately. You can understand the data at two levels **micro level** and **macro level**.
Micro Level Analysis
It pertains to an individual or a small group of individuals. For example, number of times job application submitted, number of times print this page was clicked, etc.
Macro Level Analysis
It is concerned with the primary business objectives with huge groups of people such as communities, nation, etc. For example, number of conversions in a particular demographic.
Web Analysis - What to Measure?
These are the few measurements conducted in web analytics −

- **Engagement Rate**

It shows how long a person stays on your web page. What all pages he surf. To make your web pages more engaging, include informative content, visuals, fonts and bullets.

- **Bounce Rate**

If a person leaves your website within a span of 30 sec, it is considered as a bounce. The rate at which users spin back is called the bounce rate. To minimize bounce rate include related posts, clear call-to-action and backlinks in your webpages.

- **Dashboards**

Dashboard is single page view of information important to user. You can create your own dashboards keeping in mind your requirements. You may keep only frequently viewed data on dashboard.

- **Event Tracking**

Event tracking allows you to track other activities on your website. For example, you can track downloads and sign-ups through event tracking.

- **Traffic Source**

You can overview traffic sources. You can even filter it further. Figuring out the key areas can help you learn about the area of improvement.

- **Annotations**

It allows you to view a traffic report for past time. You can click on graph and type in to save it for future study.

- **Visitor Flow**

It gives you a clear picture of pages visited and the sequence of the same. Understanding users' path may help you in re-navigation in order to give customer a hassle-free navigation.

- **Content**

It gives you insight about website's content section. You can see how each page is doing, website loading speed, etc.

- **Conversions**

Analytics lets you track goals and path used to achieve these goals. You can get details regarding, product performances, purchase amount, and mode of billing. Web Analytics offer you more than this. All you need is to analyze things minutely and keep patience.

- **Page Load Time**

More is the load time, the more is bounce rate. Tracking page load time is equally important.

- **Behavior**

Behavior lets you know page views and time spent on website. You can find out how customer behaves once he is on your website.

Big Data and Marketing

Meaning and Need

Big Data and Marketing refers to the use of large-scale customer and market data to understand customer behavior, predict demand, personalize offers, and optimize marketing campaigns. Traditional marketing decisions were based on surveys and limited sales data. Today, customers interact with brands through websites, mobile apps, social media, and digital payments, creating continuous streams of data. Big Data technologies help organizations store and analyze this information efficiently to make faster and more accurate marketing decisions.

Marketing teams use Big Data to identify what customers want, when they want it, and how they prefer to buy it. Instead of a single campaign for all users, companies create micro-targeted campaigns for specific customer groups. This improves customer satisfaction, conversion rate, and overall revenue.

Sources of Marketing Big Data

- Social media: likes, shares, comments, hashtags, influencer engagement
- Web clickstream: page views, clicks, time on page, navigation path
- Mobile app analytics: installs, session duration, in-app activity, location-based behavior
- E-commerce transactions: cart items, purchases, returns, payment methods
- CRM systems: customer profile data, support tickets, loyalty program details
- Email and messaging campaigns: open rate, click-through rate (CTR), response time
- Customer feedback: ratings, reviews, complaint text, voice-of-customer surveys

Key Marketing Analytics using Big Data

- Customer segmentation: clustering customers into groups based on demographics and behavior
- Recommendation systems: suggesting products or content based on similarity and past activity
- Customer Lifetime Value (CLV): predicting long-term revenue potential of customers
- Churn prediction: predicting which customers may stop using a service
- Sentiment analysis: finding positive/negative opinions from reviews and social media
- Campaign performance optimization: selecting best channels, timing, and ad creatives
- Dynamic pricing: adjusting prices based on demand patterns and competitor trends

Technologies and Tools

- Data storage: HDFS, cloud storage, NoSQL databases (MongoDB, Cassandra)
- Processing: Hadoop MapReduce (batch), Apache Spark (fast analytics)
- Streaming: Kafka, Spark Streaming for real-time marketing triggers

- Visualization: Tableau, Power BI, Google Data Studio
- Machine Learning: classification, clustering, regression models for predictions

Real-World Use Cases and Examples

Use Case 1: Personalized Recommendations (E-commerce)
E-commerce companies track browsing history, items added to cart, purchase frequency, and product ratings. Using Big Data analytics, the system recommends products that are most relevant to each user. For example, if a customer frequently searches for laptops and accessories, the platform suggests laptop stands, keyboards, and warranty plans. This increases cross-selling and average order value.

Use Case 2: Customer Churn Prediction (Telecom)
Telecom companies store call data records, recharge frequency, complaint history, and network usage. A churn prediction model identifies customers likely to switch to competitors. The company can then send special offers (data packs, discounts) to retain them, reducing customer loss and improving profitability.

Use Case 3: Social Media Sentiment Tracking (Brand Monitoring)
Companies use Big Data tools to monitor brand sentiment by analyzing tweets, reviews, and comments. If negative sentiment increases after a product launch, the company can quickly investigate issues and take corrective action. This helps protect reputation and improves customer trust.

Advantages and Challenges
Advantages: better targeting, higher conversion rate, improved customer satisfaction, optimized budget allocation
Challenges: privacy concerns, data quality issues, integrating multiple data sources, handling unstructured text data

In summary, Big Data makes marketing intelligent, customer-centric, and measurable. When combined with machine learning, it supports predictive and prescriptive marketing decisions suitable for modern businesses.

Fraud and Big Data

Meaning and Importance
Fraud refers to illegal or deceptive activities performed to gain financial benefits. With the growth of digital banking, online shopping, and mobile payments, fraud attempts have increased significantly. Fraud can happen at a very high speed and across multiple channels, which makes detection difficult using traditional systems. Big Data analytics enables organizations to analyze millions of transactions, detect anomalies, and stop fraud in near real-time.
Fraud detection requires analyzing not only transaction values but also additional attributes such as location, time, device information, customer history, and spending patterns. Big Data platforms process these large and fast data streams effectively and help build accurate fraud detection models.

Major Types of Fraud
- Credit card fraud: unauthorized card usage or stolen card details
- Banking fraud: phishing attacks, fake accounts, account takeover
- Insurance fraud: false claims, overbilling, staged accidents
- E-commerce fraud: fake returns, coupon misuse, chargebacks
- Identity theft: using stolen identity to access loans or services
- Money laundering: transferring illegal money through multiple layers

Fraud Detection Approaches
- Rule-based detection: predefined rules such as transaction threshold and suspicious patterns
- Anomaly detection: identifies transactions that are highly different from normal behavior

- Supervised ML: classification models trained using labeled fraud/non-fraud data
- Unsupervised learning: clustering and outlier detection when labels are not available
- Graph analytics: detects connected fraud networks and suspicious relationships

Big Data Tools for Fraud Detection

- Kafka: collects real-time transaction events from applications
- Spark Streaming / Flink: processes transactions and applies fraud models instantly
- Hadoop/HDFS: stores historical transaction logs for investigation
- NoSQL (Cassandra/HBase): stores fast-access customer behavior records
- Machine learning pipelines: build and deploy fraud scoring models

Real-World Use Cases and Examples

Use Case 1: Credit Card Transaction Fraud

A customer usually makes small purchases in a local region. If the same card is suddenly used for a large international transaction at midnight, the system considers it suspicious. A fraud detection engine checks time, location, device, and transaction history. If risk is high, the transaction is blocked and the customer is alerted for confirmation.

Use Case 2: Banking Login Fraud and Account Takeover

Attackers may attempt multiple login tries using stolen credentials. Big Data systems analyze login frequency, IP addresses, device fingerprints, and unusual time patterns. If suspicious activity is detected, the system triggers multi-factor authentication or temporarily blocks the account.

Use Case 3: Insurance Claim Fraud

Insurance companies analyze claim text, historical customer claims, hospital billing patterns, and network relationships between claimants and service providers. By using anomaly detection and clustering, repeated suspicious claims can be identified early and investigated to reduce losses.

Benefits and Challenges

Benefits: real-time detection, reduced financial losses, improved trust, better compliance
Challenges: high false positives, privacy and security requirements, evolving fraud patterns, requirement of labeled data

Overall, Big Data enables faster and smarter fraud detection by combining real-time processing with advanced analytics.

Risk and Big Data

Meaning of Risk in Business and Finance

Risk is the possibility of loss or uncertain outcomes. Organizations face risk from financial changes, customer defaults, cyber-attacks, operational failures, and reputation damage. Risk management is the practice of identifying risks, measuring their impact, and taking preventive actions to reduce losses. Big Data improves risk management by using large volumes of historical and real-time data for forecasting and early warning detection.

Types of Risk

- Market risk: losses due to changes in stock prices, interest rates, or currency rates
- Credit risk: borrower fails to repay loan
- Liquidity risk: inability to meet short-term financial obligations
- Operational risk: system failures, process errors, human mistakes
- Cyber risk: data breaches, malware, phishing attacks
- Reputational risk: loss of trust due to negative events

How Big Data Improves Risk Management
- Risk identification: detect hidden risk factors from multiple data sources
- Early warning alerts: detect unusual behavior before major loss occurs
- Continuous monitoring: track risk indicators in real-time
- Better forecasting: predictive models estimate future loss probabilities
- Stress testing: simulate extreme events such as recession or market crashes

Risk Analytics Techniques
- Predictive analytics using machine learning (classification/regression)
- Time-series forecasting for market and economic indicators
- Scenario analysis to evaluate multiple future situations
- Monte Carlo simulations for probability-based risk measurement
- Dashboards and KPIs for monitoring risk metrics

Use Cases and Examples

Use Case 1: Market Volatility Risk in Trading
Financial institutions analyze high-frequency market data, news sentiment, and economic indicators to detect volatility risk. If volatility increases beyond a threshold, risk controls such as position limits or stop-loss rules are activated to prevent large losses.

Use Case 2: Cyber Risk Detection in Banking
Banks monitor network logs, login patterns, device data, and transaction anomalies. Big Data systems detect unusual access patterns, such as many login attempts from different countries. Early detection prevents breaches and reduces operational loss.

Use Case 3: Operational Risk in E-commerce Systems
E-commerce platforms monitor server logs, payment gateway errors, and customer complaints. Big Data analytics helps detect system failures early and ensures continuity of business operations, reducing downtime and revenue loss.

Benefits and Challenges
Benefits: improved forecasting, faster response, better compliance, reduced losses
Challenges: data quality issues, high computing cost, requirement of skilled experts, privacy regulations

In summary, Big Data strengthens risk management through accurate prediction, real-time monitoring, and decision support.

Credit Risk Management

Introduction
Credit risk is the risk that a borrower will not repay a loan or credit amount as agreed. Banks and financial institutions manage credit risk to reduce loan defaults, minimize losses, and maintain healthy financial performance. Traditional credit assessment relied on limited factors such as income and credit history. Big Data introduces additional behavioral and transactional factors, enabling more accurate credit scoring and continuous monitoring.

Credit Risk Lifecycle
- Application stage: customer applies for loan/credit card
- Verification stage: KYC, employment, income, and identity verification
- Scoring stage: credit scoring and risk rating
- Decision stage: approve/reject and set interest rate/limit
- Monitoring stage: track repayments and behavior continuously
- Collection and recovery stage: manage overdue accounts and recovery actions

Key Metrics (PD, LGD, EAD)
Credit risk is measured using standard metrics used in banking and Basel regulations:
PD (Probability of Default): chance that a borrower will default within a defined time period.
LGD (Loss Given Default): percentage of money the lender may lose if default occurs after recovery.
EAD (Exposure at Default): total outstanding exposure when borrower defaults.
Expected Loss (EL) = PD × LGD × EAD, used for provisioning and risk planning.

How Big Data Improves Credit Risk Management
Uses alternative data such as digital payment behavior and utility payment patterns
Detects early warning signals like spending stress and irregular income patterns
Improves approval accuracy by reducing human bias
Supports real-time monitoring after loan approval
Helps reduce NPAs (Non-Performing Assets) by proactive interventions

Credit Risk Models and Methods
Logistic Regression: interpretable model used for PD prediction
Decision Trees: produces rule-based decisions and explains approvals
RandomForest / Gradient Boosting: better performance with ensemble learning
Neural Networks: useful for complex nonlinear patterns on large datasets
Scorecards: widely used for credit card and personal loan decisions

Use Cases and Examples

Use Case 1: Loan Approval and Pricing
Banks analyze customer income, repayment history, outstanding loans, and transaction patterns. Based on risk score, the loan can be approved with different interest rates. A low-risk customer receives lower interest rates, while high-risk borrowers may face stricter loan limits.

Use Case 2: Early Default Warning System
If a borrower starts missing minimum payments or shows sudden increase in expenses compared to income, Big Data monitoring systems generate alerts. The bank can contact the customer, offer restructuring, or reduce further exposure.

Use Case 3: Fraud Prevention in Credit Applications
Credit risk teams use Big Data to detect identity mismatch and suspicious application patterns. For example, multiple loans applied using similar address, phone number, or device fingerprint are flagged for manual verification.

Challenges in Credit Risk with Big Data
Privacy and fairness concerns in using alternative data
Bias in historical lending data
Explainability of complex models
Regulatory compliance requirements
Data integration from different banking systems

Overall, Big Data-based credit risk management improves accuracy, reduces defaults, and supports better lending decisions.

Big Data and Algorithmic Trading

Introduction
Algorithmic trading uses computer programs to automatically execute buy and sell orders based on predefined logic. Big Data is important because financial markets generate very large real-time data such as tick prices, volumes, and order books. The trading system must process this data quickly, generate

signals, and control risk. Big Data platforms enable fast data ingestion, real-time analytics, and scalable backtesting on historical market data.

Major Data Sources in Algo Trading

- Price data: OHLC and tick-by-tick prices
- Volume data: traded volume, liquidity indicators
- Order book data: bid/ask levels and market depth
- Fundamental data: earnings, financial statements, ratios
- News data: headlines, breaking news affecting prices
- Social media sentiment: public opinion signals
- Macro indicators: interest rate announcements, inflation reports

Trading Strategies Supported by Big Data

- Trend following: buy in upward trends and sell in downward trends
- Mean reversion: prices return to average; buy low and sell high
- Momentum strategies: capture strong recent movement
- Arbitrage: profit from price difference across exchanges or instruments
- High-Frequency Trading (HFT): extremely fast execution using low-latency systems
- News/sentiment trading: trade based on positive or negative sentiment

Big Data Architecture for Algo Trading

A high-level Big Data architecture for algorithmic trading includes:

- Data ingestion layer: Kafka or message queues collect market feeds
- Stream processing: Spark Streaming/Flink process real-time events
- Storage layer: HDFS/NoSQL store historical prices and logs
- Analytics & ML layer: models predict price direction and detect anomalies
- Execution layer: APIs connect to broker/exchange for placing orders
- Monitoring layer: dashboards monitor strategy performance and risk

Risk Management in Algo Trading

Stop-loss and take-profit rules
Position size limits to avoid overexposure
Volatility-based risk control (reduce trading during high volatility)
Backtesting and paper trading before live execution
Continuous monitoring to detect strategy failure or market shocks

Use Cases and Examples

Use Case 1: Trend Breakout Strategy

If the price crosses a resistance level with high volume, the algorithm triggers a buy order. If the price falls below a support level, it triggers a sell order. Big Data helps evaluate this strategy using historical tick data and real-time volume changes.

Use Case 2: Sentiment-Based Trading

If negative news sentiment increases for a company (e.g., major loss or legal issues), the algorithm reduces exposure or sells the stock. Sentiment analysis uses Big Data from news feeds and social media to generate signals.

Use Case 3: Arbitrage Trading

When the same stock or crypto asset has slightly different prices on two exchanges, the system buys on the cheaper exchange and sells on the expensive one. Big Data platforms help handle high-frequency price feeds and fast execution.

Advantages and Challenges
Advantages: speed, reduced emotional trading, ability to process huge market data, systematic risk control
Challenges: market data noise, high infrastructure cost, regulatory requirements, risk of algorithm failure

In conclusion, Big Data makes algorithmic trading scalable and efficient by enabling real-time decision-making, predictive analytics, and robust risk control.

APPLICATIONS OF BIG DATA

In today's world, there are a lot of data. Big companies utilize those data for their business growth. By analyzing this data, the useful decision can be made in various cases as discussed below:

1. Tracking Customer Spending Habit, Shopping Behavior:
In big retails store (like Amazon, Walmart, Big Bazar etc.) management team has to keep data of customer's spending habit (in which product customer spent, in which brand they wish to spent, how frequently they spent), shopping behavior, customer's most liked product (so that they can keep those products in the store). Which product is being searched/sold most, based on that data, production/collection rate of that product get fixed.
Banking sector uses their customer's spending behavior-related data so that they can provide the offer to a particular customer to buy his particular liked product by using bank's credit or debit card with discount or cashback. By this way, they can send the right offer to the right person at the right time.

2. Recommendation:
By tracking customer spending habit, shopping behavior, Big retails store provide a recommendation to the customer. E-commerce site like Amazon, Walmart, Flipkart does product recommendation. They track what product a customer is searching, based on that data they recommend that type of product to that customer.
As an example, suppose any customer searched bed cover on Amazon. So, Amazon got data that customer may be interested to buy bed cover. Next time when that customer will go to any google page, advertisement of various bed covers will be seen. Thus, advertisement of the right product to the right customer can be sent.
YouTube also shows recommend video based on user's previous liked, watched video type. Based on the content of a video, the user is watching, relevant advertisement is shown during video running. As an example suppose someone watching a tutorial video of Big data, then advertisement of some other big data course will be shown during that video.

3. Smart Traffic System:
Data about the condition of the traffic of different road, collected through camera kept beside the road, at entry and exit point of the city, GPS device placed in the vehicle (Ola, Uber cab, etc.). All such data are analyzed and jam-free or less jam way, less time taking ways are recommended. Such a way smart traffic system can be built in the city by Big data analysis. One more profit is fuel consumption can be reduced.

4. Secure Air Traffic System:
At various places of flight (like propeller etc) sensors present. These sensors capture data like the speed of flight, moisture, temperature, other environmental condition. Based on such data analysis, an environmental parameter within flight are set up and varied.
By analyzing flight's machine-generated data, it can be estimated how long the machine can operate flawlessly when it to be replaced/repaired.

5. Auto Driving Car:
Big data analysis helps drive a car without human interpretation. In the various spot of car camera, a sensor placed, that gather data like the size of the surrounding car, obstacle, distance from those, etc.

These data are being analyzed, then various calculation like how many angles to rotate, what should be speed, when to stop, etc carried out. These calculations help to take action automatically.

6. Virtual Personal Assistant Tool:

Big data analysis helps virtual personal assistant tool (like Siri in Apple Device, Cortana in Windows, Google Assistant in Android) to provide the answer of the various question asked by users. This tool tracks the location of the user, their local time, season, other data related to question asked, etc. Analyzing all such data, it provides an answer.

As an example, suppose one user asks "Do I need to take Umbrella?", the tool collects data like location of the user, season and weather condition at that location, then analyze these data to conclude if there is a chance of raining, then provide the answer.

7. IoT:

Manufacturing company install IOT sensor into machines to collect operational data. Analyzing such data, it can be predicted how long machine will work without any problem when it requires repairing so that company can take action before the situation when machine facing a lot of issues or gets totally down. Thus, the cost to replace the whole machine can be saved.

In the Healthcare field, Big data is providing a significant contribution. Using big data tool, data regarding patient experience is collected and is used by doctors to give better treatment. IoT device can sense a symptom of probable coming disease in the human body and prevent it from giving advance treatment. IoT Sensor placed near-patient, new-born baby constantly keeps track of various health condition like heart bit rate, blood presser, etc. Whenever any parameter crosses the safe limit, an alarm sent to a doctor, so that they can take step remotely very soon.

8. Education Sector:

Online educational course conducting organization utilize big data to search candidate, interested in that course. If someone searches for YouTube tutorial video on a subject, then online or offline course provider organization on that subject send ad online to that person about their course.

9. Energy Sector:

Smart electric meter read consumed power every 15 minutes and sends this read data to the server, where data analyzed and it can be estimated what is the time in a day when the power load is less throughout the city. By this system manufacturing unit or housekeeper are suggested the time when they should drive their heavy machine in the night time when power load less to enjoy less electricity bill.

10. Media and Entertainment Sector:

Media and entertainment service providing company like Netflix, Amazon Prime, Spotify do analysis on data collected from their users. Data like what type of video, music users are watching, listening most, how long users are spending on site, etc are collected and analyzed to set the next business strategy.

BIG DATA TECHNOLOGIES

Big data technologies can be categorized into four main types: data storage, data mining, data analytics, and data visualization [2]. Each of these is associated with certain tools, and you'll want to choose the right tool for your business needs depending on the type of big data technology required.

1. Data storage

Big data technology that deals with data storage has the capability to fetch, store, and manage big data. It is made up of infrastructure that allows users to store the data so that it is convenient to access. Most data storage platforms are compatible with other programs. Two commonly used tools are Apache Hadoop and MongoDB.

- **Apache Hadoop:** Apache is the most widely used big data tool. It is an open- source software platform that stores and processes big data in a distributed computing environment across hardware clusters. This distribution allows for faster data processing. The framework is designed to reduce bugs or faults, be scalable, and process all data formats.

- **MongoDB:** MongoDB is a NoSQL database that can be used to store large volumes of data. Using key-value pairs (a basic unit of data), MongoDB categorizes documents into collections. It is written in C, C++, and JavaScript, and is one of the most popular big data databases because it can manage and store unstructured data with ease.

2. Data mining

Data mining extracts the useful patterns and trends from the raw data. Big data technologies such as Rapidminer and Presto can turn unstructured and structured data into usable information.

- **Rapidminer:** Rapidminer is a data mining tool that can be used to build predictive models. It draws on these two roles as strengths, of processing and preparing data, and building machine and deep learning models. The end-to- end model allows for both functions to drive impact across the organization.
- **Presto:** Presto is an open-source query engine that was originally developed by Facebook to run analytic queries against their large datasets. Now, it is available widely. One query on Presto can combine data from multiple sources within an organization and perform analytics on them in a matter of minutes.

3. Data analytics

In big data analytics, technologies are used to clean and transform data into information that can be used to drive business decisions. This next step (after data mining) is where users perform algorithms, models, and predictive analytics using tools such as Apache Spark and Splunk.

- **Apache Spark:** Spark is a popular big data tool for data analysis because it is fast and efficient at running applications. It is faster than Hadoop because it uses random access memory (RAM) instead of being stored and processed in batches via MapReduce . Spark supports a wide variety of data analytics tasks and queries.
- **Splunk:** Splunk is another popular big data analytics tool for deriving insights from large datasets. It has the ability to generate graphs, charts, reports, and dashboards. Splunk also enables users to incorporate artificial intelligence (AI) into data outcomes.

4. Data visualization

Finally, big data technologies can be used to create stunning visualizations from the data. In data-oriented roles, data visualization is a skill that is beneficial for presenting recommendations to stakeholders for business profitability and operations—to tell an impactful story with a simple graph.

- **Tableau:** Tableau is a very popular tool in data visualization because its drag- and-drop interface makes it easy to create pie charts, bar charts, box plots, Gantt charts, and more. It is a secure platform that allows users to share visualizations and dashboards in real time.
- **Looker:** Looker is a business intelligence (BI) tool used to make sense of big data analytics and then share those insights with other teams. Charts, graphs, and dashboards can be configured with a query, such as monitoring weekly brand engagement through social media analytics.

OPEN SOURCE TECHNOLOGIES / BIG DATA ANALYTICS TOOLS

There are hundreds of data analytics tools out there in the market today but the selection of the right tool will depend upon your business NEED, GOALS, and VARIETY to get business in the right direction. Now, let's check out the top 10 analytics tools in big data.

1. APACHE Hadoop

It's a Java-based open-source platform that is being used to store and process big data. It is built on a cluster system that allows the system to process data efficiently and let the data run parallel. It can process both structured and unstructured data from one server to multiple computers. Hadoop also offers cross-platform support for its users. Today, it is the best big data analytic tool and is popularly used by many tech giants such as Amazon, Microsoft, IBM, etc.

Features of Apache Hadoop:

- Free to use and offers an efficient storage solution for businesses.
- Offers quick access via HDFS (Hadoop Distributed File System).
- Highly flexible and can be easily implemented with MySQL, and JSON.
- Highly scalable as it can distribute a large amount of data in small segments.
- It works on small commodity hardware like JBOD or a bunch of disks.

2. Cassandra

APACHE Cassandra is an open-source NoSQL distributed database that is used to fetch large amounts of data. It's one of the most popular tools for data analytics and has been praised by many tech companies due to its high scalability and availability without compromising speed and performance. It is capable of delivering thousands of operations every second and can handle petabytes of resources with almost zero downtime. It was created by Facebook back in 2008 and was published publicly.

Features of APACHE Cassandra:

- **Data Storage Flexibility:** It supports all forms of data i.e. structured, unstructured, semi-structured, and allows users to change as per their needs.
- Data Distribution System: Easy to distribute data with the help of replicating data on multiple data centers.
- **Fast Processing:** Cassandra has been designed to run on efficient commodity hardware and also offers fast storage and data processing.
- **Fault-tolerance:** The moment, if any node fails, it will be replaced without any delay.

3. Qubole

It's an open-source big data tool that helps in fetching data in a value of chain using ad-hoc analysis in machine learning. Qubole is a data lake platform that offers end-to-end service with reduced time and effort which are required in moving data pipelines. It is capable of configuring multi-cloud services such as AWS, Azure, and Google Cloud. Besides, it also helps in lowering the cost of cloud computing by 50%.

Features of Qubole:

- **Supports ETL process:** It allows companies to migrate data from multiple sources in one place.
- **Real-time Insight**: It monitors user's systems and allows them to view real-time insights
- **Predictive Analysis**: Qubole offers predictive analysis so that companies can take actions accordingly for targeting more acquisitions.
- **Advanced Security System:** To protect users' data in the cloud, Qubole uses an advanced security system and also ensures to protect any future breaches. Besides, it also allows encrypting cloud data from any potential threat.

4. Xplenty

It is a data analytic tool for building a data pipeline by using minimal codes in it. It offers a wide range of solutions for sales, marketing, and support. With the help of its interactive graphical interface, it provides solutions for ETL, ELT, etc. The best part of using Xplenty is its low investment in hardware & software and its offers support via email, chat, telephonic and virtual meetings. Xplenty is a platform to process data for analytics over the cloud and segregates all the data together.

Features of Xplenty:

- **Rest API:** A user can possibly do anything by implementing Rest API
- **Flexibility:** Data can be sent, and pulled to databases, warehouses, and salesforce.
- **Data Security:** It offers SSL/TSL encryption and the platform is capable of verifying algorithms and certificates regularly.
- **Deployment:** It offers integration apps for both cloud & in-house and supports deployment to integrate apps over the cloud.

5. Spark

APACHE Spark is another framework that is used to process data and perform numerous tasks on a large scale. It is also used to process data via multiple computers with the help of distributing tools. It is widely used among data analysts as it offers easy-to-use APIs that provide easy data pulling methods and it is capable of handling multi-petabytes of data as well. Recently, Spark made a record of processing 100 terabytes of data in just 23 minutes which broke the previous world record of Hadoop (71 minutes). This is the reason why big tech giants are moving towards spark now and is highly suitable for ML and AI today.

Features of APACHE Spark:

- **Ease of use:** It allows users to run in their preferred language. (JAVA, Python, etc.)
- **Real-time Processing:** Spark can handle real-time streaming via Spark Streaming
- **Flexible:** It can run on, Mesos, Kubernetes, or the cloud.

6. Mongo DB

Came in limelight in 2010, is a free, open-source platform and a document- oriented (NoSQL) database that is used to store a high volume of data. It uses collections and documents for storage and its document consists of key-value pairs which are considered a basic unit of Mongo DB. It is so popular among developers due to its availability for multi-programming languages such as Python, Jscript, and Ruby.

Features of Mongo DB:

- **Written in C++:** It's a schema-less DB and can hold varieties of documents inside.
- **Simplifies Stack:** With the help of mongo, a user can easily store files without any disturbance in the stack.
- **Master-Slave Replication:** It can write/read data from the master and can be called back for backup.

7. Apache Storm

A storm is a robust, user-friendly tool used for data analytics, especially in small companies. The best part about the storm is that it has no language barrier (programming) in it and can support any of them. It was designed to handle a pool of large data in fault-tolerance and horizontally scalable methods. When we talk about real-time data processing, Storm leads the chart because of its distributed real-time big data processing system, due to which today many tech giants are using APACHE Storm in their system. Some of the most notable names are Twitter, Zendesk, NaviSite, etc.

Features of Storm:

- **Data Processing:** Storm process the data even if the node gets disconnected
- **Highly Scalable:** It keeps the momentum of performance even if the load increases
- **Fast:** The speed of APACHE Storm is impeccable and can process up to 1 million messages of 100 bytes on a single node.

8. SAS

Today it is one of the best tools for creating statistical modeling used by data analysts. By using SAS, a data scientist can mine, manage, extract or update data in different variants from different sources. Statistical Analytical System or SAS allows a user to access the data in any format (SAS tables or Excel worksheets). Besides that it also offers a cloud platform for business analytics called SAS Viya and also to get a strong grip on AI & ML, they have introduced new tools and products.

Features of SAS:

- **Flexible Programming Language:** It offers easy-to-learn syntax and has also vast libraries which make it suitable for non-programmers
- **Vast Data Format:** It provides support for many programming languages which also include SQL and carries the ability to read data from any format.
- **Encryption:** It provides end-to-end security with a feature called SAS/SECURE.

9. Data Pine

Datapine is an analytical used for BI and was founded back in 2012 (Berlin, Germany). In a short period of time, it has gained much popularity in a number of countries and it's mainly used for data extraction (for small-medium companies fetching data for close monitoring). With the help of its enhanced UI design, anyone can visit and check the data as per their requirement and offer in 4 different price brackets, starting from $249 per month. They do offer dashboards by functions, industry, and platform.

Features of Datapine:

- **Automation:** To cut down the manual chase, datapine offers a wide array of AI assistant and BI tools.
- **Predictive Tool:** datapine provides forecasting/predictive analytics by using historical and current data, it derives the future outcome.
- **Add on:** It also offers intuitive widgets, visual analytics & discovery, ad hoc reporting, etc.

10. Rapid Miner

It's a fully automated visual workflow design tool used for data analytics. It's a no-code platform and users aren't required to code for segregating data. Today, it is being heavily used in many industries such as ed-tech, training, research, etc. Though it's an open-source platform but has a limitation of adding 10000 data rows and a single logical processor. With the help of Rapid Miner, one can easily deploy their ML models to the web or mobile (only when the user interface is ready to collect real-time figures).

Features of Rapid Miner:

- **Accessibility:** It allows users to access 40+ types of files (SAS, ARFF, etc.) via URL
- **Storage:** Users can access cloud storage facilities such as AWS and dropbox
- **Data validation:** Rapid miner enables the visual display of multiple results in history for better evaluation.

CLOUD AND BIG DATA

2. Big Data:

Big data refers to the data which is huge in size and also increasing rapidly with respect to time. Big data includes structured data, unstructured data as well as semi-structured data. Big data cannot be stored and processed in traditional data management tools it needs specialized big data management tools. It refers to complex and large data sets having 5 V's volume, velocity, Veracity, Value and variety information assets. It includes data storage, data analysis, data mining and data visualization.

Examples of the sources where big data is generated includes social media data, e- commerce data, weather station data, IoT Sensor data etc.

Characteristics of Big Data :

- Variety of Big data – Structured, unstructured, and semi structured data
- Velocity of Big data – Speed of data generation
- Volume of Big data – Huge volumes of data that is being generated
- Value of Big data – Extracting useful information and making it valuable
- Variability of Big data – Inconsistency which can be shown by the data at times.

Advantages of Big Data :

- Cost Savings
- Better decision-making
- Better Sales insights
- Increased Productivity
- Improved customer service.

Disadvantages of Big Data :

- Incompatible tools

- Security and Privacy Concerns
- Need for cultural change
- Rapid change in technology
- Specific hardware needs.

3. Cloud Computing :

Cloud computing refers to the on demand availability of computing resources over internet. These resources includes servers, storage, databases, software, analytics, networking and intelligence over the Internet and all these resources can be used as per requirement of the customer. In cloud computing customers have to pay as per use. It is very flexible and can be resources can be scaled easily depending upon the requirement. Instead of buying any IT resources physically, all resources can be availed depending on the requirement from the cloud vendors. Cloud computing has three service models i.e Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS).
Examples of cloud computing vendors who provides cloud computing services are Amazon Web Service (AWS), Microsoft Azure, Google Cloud Platform, IBM Cloud Services etc.

Characteristics of Cloud Computing :

- On-Demand availability
- Accessible through a network
- Elastic Scalability
- Pay as you go model
- Multi-tenancy and resource pooling.

Advantages of Cloud Computing :

- Back-up and restore data
- Improved collaboration
- Excellent accessibility
- Low maintenance cost
- On-Demand Self-service.

Disadvantages of Cloud Computing:

- Vendor lock-in
- Limited Control
- Security Concern
- Downtime due to various reason
- Requires good Internet connectivity.

Difference between Big Data and Cloud Computing:

S.No.	BIG DATA	CLOUD COMPUTING
01.	Big data refers to the data which is huge in size and also increasing rapidly with respect to time.	Cloud computing refers to the on demand availability of computing resources over internet.
02.	Big data includes structured data, unstructured data as well as semi- structured data.	Cloud Computing Services includes Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS).

03.	Volume of data, Velocity of data, Variety of data, Veracity of data, and Value of data are considered as the 5 most important characteristics of Big data.	On-Demand availability of IT resources, broad network access, resource pooling, elasticity and measured service are considered as the main characteristics of cloud computing.
04.	The purpose of big data is to organizing the large volume of data and extracting the useful information from it and using that information for the improvement of business.	The purpose of cloud computing is to store and process data in cloud or availing remote IT services without physically installing any IT resources.
05.	Distributed computing is used for analyzing the data and extracting the useful information.	Internet is used to get the cloud based services from different cloud vendors.
06.	Big data management allows centralized platform, provision for backup and recovery and low maintenance cost.	Cloud computing services are cost effective, scalable and robust.
07.	Some of the challenges of big data are variety of data, data storage and integration, data processing and resource management.	Some of the challenges of cloud computing are availability, transformation, security concern, charging model.
08.	Big data refers to huge volume of data, its management, and useful information extraction.	Cloud computing refers to remote IT resources and different internet service models.
09.	Big data is used to describe huge volume of data and information.	Cloud computing is used to store data and information on remote servers and also processing the data using remote infrastructure.
10.	Some of the sources where big data is generated includes social media data, e-commerce data, weather station data, IoT Sensor data etc.	Some of the cloud computing vendors who provides cloud computing services are Amazon Web Service (AWS), Microsoft Azure, Google Cloud Platform, IBM Cloud Services etc.

MOBILE BUSINESS INTELLIGENCE

Business Intelligence

"Business Intelligence is not just about turning data into information, rather organizations need that data to impact how their business operates and responds to the changing marketplace."

So, it is not all about transforming data into information, though Business Intelligence significantly involves this process. Business Intelligence is transforming data into meaningful, actionable insights that enable organizations to make informed business strategies and tactical decisions.

Mobile Business Intelligence

Business Intelligence delivers relevant and trustworthy information to the right person at the right time. Mobile business intelligence is the transfer of business intelligence from the desktop to mobile devices such as the BlackBerry, iPad, and iPhone.

The ability to access analytics and data on mobile devices or tablets rather than desktop computers is referred to as mobile business intelligence. The business metric dashboard and key performance indicators (KPIs) are more clearly displayed.

With the rising use of mobile devices, so have the technology that we all utilise in our daily lives to make our lives easier, including business. Many businesses have benefited from mobile business intelligence. Essentially, this post is a guide for business owners and others to educate them on the benefits and pitfalls of Mobile BI.

Need for mobile BI?

Mobile phones' data storage capacity has grown in tandem with their use. You are expected to make decisions and act quickly in this fast-paced environment. The number of businesses receiving assistance in such a situation is growing by the day.

To expand your business or boost your business productivity, mobile BI can help, and it works with both small and large businesses. Mobile BI can help you whether you are a salesperson or a CEO. There is a high demand for mobile BI in order to reduce information time and use that time for quick decision making.

As a result, timely decision-making can boost customer satisfaction and improve an enterprise's reputation among its customers. It also aids in making quick decisions in the face of emerging risks.

Data analytics and visualisation techniques are essential skills for any team that wants to organise work, develop new project proposals, or wow clients with impressive presentations.

Advantages of mobile BI

1. **Simple access**

Mobile BI is not restricted to a single mobile device or a certain place. You can view your data at any time and from any location. Having real-time visibility into a firm improves production and the daily efficiency of the business. Obtaining a company's perspective with a single click simplifies the process.

2. **Competitive advantage**

Many firms are seeking better and more responsive methods to do business in order to stay ahead of the competition. Easy access to real-time data improves company opportunities and raises sales and capital. This also aids in making the necessary decisions as market conditions change.

3. **Simple decision-making**

As previously stated, mobile BI provides access to real-time data at any time and from any location. During its demand, Mobile BI offers the information. This assists consumers in obtaining what they require at the time. As a result, decisions are made quickly.

4. **Increase Productivity**

By extending BI to mobile, the organization's teams can access critical company data when they need it. Obtaining all of the corporate data with a single click frees up a significant amount of time to focus on the smooth and efficient operation of the firm. Increased productivity results in a smooth and quick-running firm.

Disadvantages of mobile

1. **Stack of data**

The primary function of a mobile BI is to store data in a systematic manner and then present it to the user as required. As a result, Mobile BI stores all of the information and does end up with heaps of earlier data. The corporation only needs a small portion of the previous data, but they need to store the entire information, which ends up in the stack

2. **Expensive**

Mobile BI can be quite costly at times. Large corporations can continue to pay for their expensive services, but small businesses cannot. As the cost of mobile BI is not sufficient, we must additionally consider the rates of IT workers for the smooth operation of BI, as well as the hardware costs involved.

However, larger corporations do not settle for just one Mobile BI provider for their organisations; they require multiple. Even when doing basic commercial transactions, mobile BI is costly.

3 **Time consuming**

Businesses prefer Mobile BI since it is a quick procedure. Companies are not patient enough to wait for data before implementing it. In today's fast-paced environment, anything that can produce results quickly is valuable. The data from the warehouse is used to create the system, hence the implementation of BI in an enterprise takes more than 18 months.

4 Data breach

The biggest issue of the user when providing data to Mobile BI is data leakage. If you handle sensitive data through Mobile BI, a single error can destroy your data as well as make it public, which can be detrimental to your business.

Many Mobile BI providers are working to make it 100 percent secure to protect their potential users' data. It is not only something that mobile BI carriers must consider, but it is also something that we, as users, must consider when granting data access authorization.

5 Poor quality data

Because we work online in every aspect, we have a lot of data stored in Mobile BI, which might be a significant problem. This means that a large portion of the data analysed by Mobile BI is irrelevant or completely useless. This can speed down the entire procedure. This requires you to select the data that is important and may be required in the future.

Best Mobile BI tools

1. **Si Sense**

Sisense is a flexible business intelligence (BI) solution that includes powerful analytics, visualisations, and reporting capabilities for managing and supporting corporate data. Businesses can use the solution to evaluate large, diverse databases and generate relevant business insights. You may easily view enormous volumes of complex data with Si Sense's code-first, low-code, and even no-code technologies. Si Sense was established in 2004 with its headquarters in New York.

Since then, the team has only taken precautionary steps in their investigation. Once the company had received $ 4 million in funding from investors, they began to pace its research.

2 **SAP Roambi analytics**

Roambi analytics is a BI tool that offers a solution that allows you to fundamentally rethink your data analysis, making it easier and faster while also increasing your data interaction.

You can consolidate all of your company's data in a single tool using SAP Roambi Analytics, which integrates all ongoing systems and data. Use of SAP Roambi analysis is a simple three-step technique. Upload your html or spreadsheet files first. The information is subsequently transformed into informative data or graphs, as well as data that may be visualised.

After the data is collected, you may easily share it with your preferred device. Roambi Analytics was founded in 2008 by a team based in California.

3 **Microsoft Power BI pro**

Microsoft's strength BI is an easy-to-use tool for all non-technical business owners. who are unfamiliar with BI tools but wish to aggregate, analyse, visualise, and share data you only need a basic understanding of Excel and other Microsoft tools, and if you are familiar with these, the Microsoft BI tool can be used as a self-service tool. Microsoft Power BI has a unique feature that allows users to create subsets of data and then automatically apply analytics to that information.

4 **IBM Cognos Analytics**

Cognos Analytics is an IBM-registered web-based business intelligence tool. Cognos Analytics is now merging with Watsons, and the benefits for users are extremely exciting. Watson cognos analytics will assist in connecting and cleaning the users' data, resulting in proper visualised data.

That way, the business owner will know where they stand in comparison to their competitors and where they can grow in the future. It combines reporting, modelling, analysis, dashboards to help you understand your organization's data and make sound business decisions.

5 Amazon quick sights

Amazon Quick View assists in the creation and distribution of interactive BI dashboards to their users,

as well as the retrieval of answers in natural language queries in seconds. Quick sight can be accessed through any device embedded in any website, portal, or app.
Amazon Quick Sight allows you to quickly and easily create interactive dashboards and reports for your users. Anyone in your organisation can securely access those dashboards via browsers or mobile devices.
Quick sight's eye-catching feature is its pay-per-session model, which allows users to use the creative dashboard created by another without paying much. The user pays according to the length of the session, with prices ranging from $0.30 for a 30-minute session to $5 for unlimited use per month per user.

CROWD SOURCING ANALYTICS

Crowdsourcing is a sourcing model in which an individual or an organization gets support from a large, open-minded, and rapidly evolving group of people in the form of ideas, micro-tasks, finances, etc. Crowdsourcing typically involves the use of the internet to attract a large group of people to divide tasks or to achieve a target. The term was coined in 2005 by Jeff Howe and Mark Robinson. Crowdsourcing can help different types of organizations get new ideas and solutions, deeper consumer engagement, optimization of tasks, and several other things.
Let us understand this term deeply with the help of an example. Like GeeksforGeeks is giving young minds an opportunity to share their knowledge with the world by contributing articles, videos of their respective domain. Here GeeksforGeeks is using the crowd as a source not only to expand their community but also to include ideas of several young minds improving the quality of the content.

Where Can We Use Crowdsourcing?

Crowdsourcing is touching almost all sectors from education to health. It is not only accelerating innovation but democratizing problem-solving methods. Some fields where crowdsourcing can be used.

1) Enterprise
2) IT
3) Marketing
4) Education
5) Finance
6) Science and Health

How to Crowdsource?

1. For scientific problem solving, a broadcast search is used where an organization mobilizes a crowd to come up with a solution to a problem.
2. For information management problems, knowledge discovery and management is used to find and assemble information.
3. For processing large datasets, distributed human intelligence is used. The organization mobilizes a crowd to process and analyze the information.

Examples of Crowdsourcing

1. **Doritos**: It is one of the companies which is taking advantage of crowdsourcing for a long time for an advertising initiative. They use consumer-created ads for one of their 30-Second Super Bowl Spots(Championship Game of Football).
2. **Starbucks**: Another big venture which used crowdsourcing as a medium for idea generation. Their white cup contest is a famous contest in which customers need to decorate their Starbucks cup with an original design and then take a photo and submit it on social media.
3. **Lays**:” Do us a flavor” contest of Lays used crowdsourcing as an idea-generating medium. They asked the customers to submit their opinion about the next chip flavor they want.
4. **Airbnb**: A very famous travel website that offers people to rent their houses or apartments by listing them on the website. All the listings are crowdsourced by people.

Crowdsourced Marketing

As discussed already crowdsourcing helps grow businesses grow a lot. May it be a business idea or just a logo design, crowdsourcing engages people directly and in turn, saves money and energy. In the upcoming years, crowdsourced marketing will surely get a boost as the world is accepting technology faster.

Main Types of Crowdsourcing

Crowdsourcing involves obtaining information or resources from a wide swath of people. In general, we can break this up into four main categories:

- **Wisdom** - Wisdom of crowds is the idea that large groups of people are collectively smarter than individual experts when it comes to problem-solving or identifying values (like the weight of a cow or number of jelly beans in a jar).
- **Creation** - Crowd creation is a collaborative effort to design or build something. Wikipedia and other wikis are examples of this. Open-source software is another good example.
- **Voting** - Crowd voting uses the democratic principle to choose a particular policy or course of action by "polling the audience."
- **Funding** - Crowdfunding involved raising money for various purposes by soliciting relatively small amounts from a large number of funders.

Crowdsourcing Sites

Here is the list of some famous crowdsourcing and crowdfunding sites.

1) Kickstarter
2) GoFundMe
3) Patreon
4) RocketHub

Advantages of Crowdsourcing

1) Evolving Innovation: Innovation is required everywhere and in this advancing world innovation has a big role to play. Crowdsourcing helps in getting innovative ideas from people belonging to different fields and thus helping businesses grow in every field.
2) Save costs: There is the elimination of wastage of time of meeting people and convincing them. Only the business idea is to be proposed on the internet and you will be flooded with suggestions from the crowd.
3) Increased Efficiency: Crowdsourcing has increased the efficiency of business models as several expertise ideas are also funded.

Disadvantages of Crowdsourcing

1) Lack of confidentiality: Asking for suggestions from a large group of people can bring the threat of idea stealing by other organizations.
2) Repeated ideas: Often contestants in crowdsourcing competitions submit repeated, plagiarized ideas which leads to time wastage as reviewing the same ideas is not worthy.

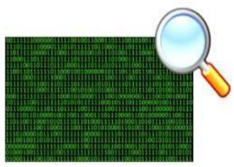
INTER AND TRANS FIREWALL ANALYTICS

Introduction

- As consumer demand is growing more and more as each day passes so to keep up with the demand companies are trying to find new ways to improve their production and market understanding mainly through firewall analytics.
- Firewall analytics is a type of data analytics which unlike as the name would suggest is not actually about analysis of the firewall but rather on all the data that the firewall encloses.
- Companies perform firewall analysis to get a better understanding of their companies functionality, supply chain, market position and their competitor's position.

Introduction

- There are two types of firewall analytics which are most widely used are as follows
  - **Intra-firewall analytics**
  - **Trans-firewall analytics**



- The general trend for a company is to first perform intra-firewall analytics and only then perform trans-firewall analytics.

or

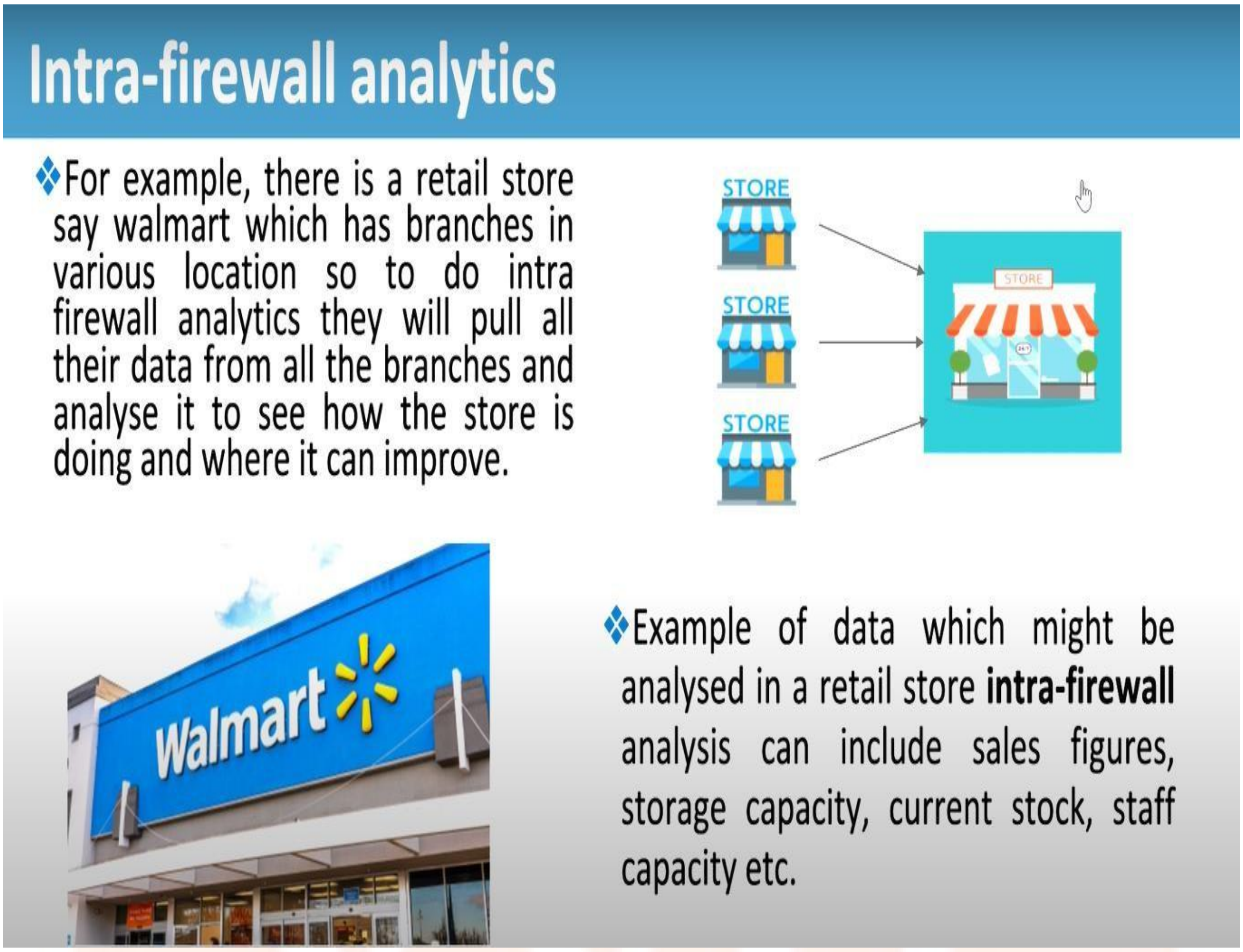
- Today companies are doing intra-firewall analytics so tomorrow they can perform trans-firewall analytics .

Intra-firewall analytics

- When companies analyse their own company's data or the data within their firewall it is called **Intra-firewall analytics**.
- It is the most widely used firewall analytics in the industries.
- In this analytics the information to noise ratio is very large as there is no interaction between the firewall and the public domain.
- Information to noise ratio is the ratio of actual useful data for the analytics to basically everything else.
- In firewall analytics information to noise ratio should be large.

Intra-firewall analytics

- The technological requirements is very less as all the data being analysed is already inside the firewall.
- The analytical method requirement is also very less as all data is from within the firewall
- It is cheaper to do as less resources are required.
- The company's privacy is maintained as there is no sharing of data outside the firewall.



Inter-firewall analytics

- **Focus**: Analyzes traffic flows between different firewalls within a network.
- **Methodology**: Utilizes data collected from multiple firewalls to identify anomalies and potential breaches.
- **Benefits**: Provides a comprehensive view of network traffic flow and helps identify lateral movement across different security zones.
- **Limitations**: Requires deployment of multiple firewalls within the network and efficient data exchange mechanisms between them.

Trans-firewall analytics

- When companies analyse their own data with the data outside their firewall i.e the data available in the public domain its called **trans-firewall analytics**.
- The public data can include data of their competitor to get an edge on them or data related to the supply chain to get a better understanding of the market.
- For trans-firewall analytics to happen intra-firewall is must.

Trans-firewall analytics

- In this analytics the information to noise ratio decreases drastically. It is because the data is taken from the public domain and is not yet filtered.
- The technological requirements are higher because the data to be used is outside the firewall.
- Similarly analytical requirements also increases for tran-firewall analytics.

Trans-firewall analytics

- No sensitive data is shared between the companies, only the publicly available data is used.

- It is more expensive than intra-firewall analytics as more resources are required.

- It is also harder to do as multiple data analyst are required to filter the information from the noise.

Trans-firewall analytics

- For example, Let us consider the 2 prestigious space agencies ISRO and NASA

- When we analyse and compare the mars mission of both the organizations like their budget, time-taken, man-power used and success rate of their respective missions.Then this type of analysis comes under the **Trans-firewall analysis**.

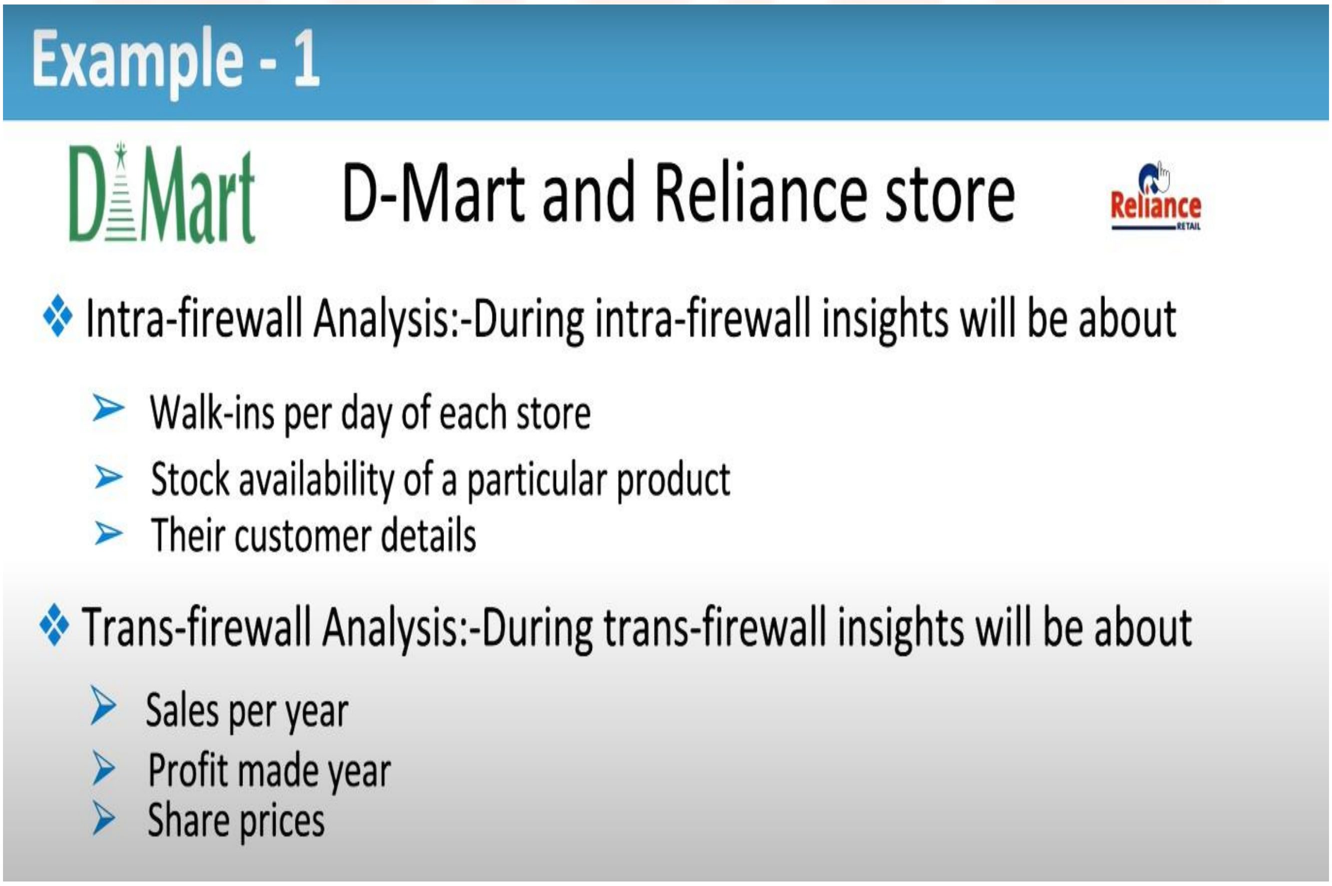
Trans-firewall analytics

- **Focus**: Analyzes encrypted traffic that traverses firewalls, which traditional security solutions may not be able to decrypt and inspect.
- **Methodology**: Uses deep packet inspection (DPI) and other advanced techniques to analyze the content of encrypted traffic without compromising its security.
- **Benefits**: Provides insight into previously hidden threats within encrypted traffic and helps detect sophisticated attacks.
- **Limitations**: Requires specialized hardware and software solutions for DPI, and raises concerns regarding potential data privacy violations.

Choosing the right approach

The choice between inter-firewall and trans-firewall analytics depends on several factors, including:

- **Network size and complexity**: Larger and more complex networks benefit more from inter-firewall analytics for comprehensive monitoring.
- **Security needs and threats**: Trans-firewall analytics is crucial for networks handling sensitive data and facing advanced threats.
- **Budget and resources**: Implementing trans-firewall analytics requires additional investment in specialized hardware and software.



Conclusion

Intra-firewall analytics	Trans-firewall analytics
Performs on one's own data or data that is enclosed by one's own firewall.	Performs on one's own data as well as data present in the public domain.
Is more widely used.	Is less used in comparison.
Information to noise ratio is more.	Information to noise ratio is very less.
Technological requirement is very less.	Special technology is required for this kind of analytics.
Analytical methods requirement is also very less.	Special analytical methods are required for this analytics
It is more cheaper to do.	It is expensive to do.
It is much more easier to do.	It is much harder to do in comparison.

---End of Unit-1 Notes---